

Session 11: EDA — Covariation — Pen-and-Paper Pair Exercise

PSY 410 | Data Science for Psychology

Name: _____ Date: _____

No laptop today? Work with a partner who has one and compare answers at the end.

The data: therapy_data (N = 150)

Post-treatment depression (BDI-II) for 50 participants in each of three conditions. *Lower scores = less depression.*

condition	n	mean	sd	min	max
Control	50	18.0	5.0	6	30
CBT	50	12.0	5.0	1	24
Mindfulness	50	14.0	5.0	3	26

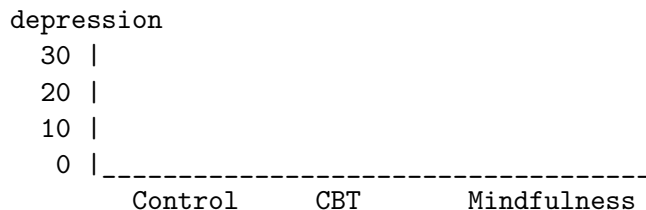
The task (same as the slide exercise)

Build a **raincloud plot** of `depression_post` by `condition` using `ggrain::geom_rain()`.

Step 1: Recall the layers. A raincloud combines three views of the same data:

1. _____ — distribution shape
2. _____ — 5-number summary
3. _____ — individual data points

Step 2: Sketch it. Using the summary stats above, draw all three layers for each condition (half-density on top, boxplot middle, jittered dots underneath). Mark each group's **mean** with an "X".



Which condition has the lowest median? _____

Step 3: Fill in the code.

```
library(ggplot2)

ggplot(therapy_data, aes(x = _____, y = _____, fill = _____)) +
  geom______(alpha = 0.6) +
  labs(title = "_____",
        x = NULL, y = "Depression score (BDI-II)") +
  theme_minimal()
```

Step 4: Why this plot? Name two things a raincloud shows that a boxplot does not:

1. _____
2. _____

Check your work

Compare your sketch and code with your partner's screen. Does your raincloud show the same overall pattern (CBT lowest, Control highest)?